DATABASE MANAGEMENT SYSTEM LAB

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LAB EXPERIMENT – 04

SQL User Management & Access Control

Title : Implementation of User Creation, Deletion, and Privilege Management in SQL

Problem Statement

Design and execute SQL commands to create users, manage their access privileges, and remove users in a database system. This assignment focuses on database security and access control mechanisms.

Step 1: Create Database & Tables

```sql
CREATE DATABASE college_db;
USE college_db;
CREATE TABLE Student (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    age INT
);
CREATE TABLE Course (
    course_id INT PRIMARY KEY AUTO_INCREMENT,
    course_name VARCHAR(50)
);
```

Tasks :

Part A: User Creation

1. **Create a new user named user1 with a password.**

```sql
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password1';
```

2. **Create another user user2 with limited access.**

```sql
CREATE USER 'user2'@'localhost' IDENTIFIED BY 'password2';
```

Part B: Granting Privileges

3. Grant SELECT and INSERT privileges on a table Student to user1.

```
GRANT SELECT, INSERT ON LabSheet4_DB.Student TO 'user1'@'localhost';
```

4. Grant ALL privileges on a table Course to user2.

```
GRANT ALL PRIVILEGES ON LabSheet4_DB.Course TO 'user2'@'localhost';
```

5. Provide WITH GRANT OPTION to user1 and observe its effect.

```
GRANT SELECT, INSERT ON LabSheet4_DB.Student
TO 'user1'@'localhost'
WITH GRANT OPTION;
```

OUTPUT:

Result Grid | Filter Rows:

user	host
departments_user	localhost
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost
root	localhost
user1	localhost
user2	localhost

Part C: Revoking Privileges

6. Revoke INSERT permission from user1 on the Student table.

```
REVOKE INSERT ON LabSheet4_DB.Student FROM 'user1'@'localhost';
```

7. Revoke all privileges from user2 on the Course table.

```
REVOKE ALL PRIVILEGES ON LabSheet4_DB.Course FROM 'user2'@'localhost';
```

Part D: User Deletion

8. Drop/Delete user user2 from the database.

```
DROP USER 'user2'@'localhost';
```

9. Attempt to login using deleted credentials and observe the result.

```
31 •  SELECT user, host FROM mysql.user WHERE user = 'user2';
```

OUTPUT:

Result Grid

	user	host

Part E: Additional Commands

10. Display all users in the system.

```
33 •  SELECT user, host FROM mysql.user;
```

OUTPUT:

Result Grid Filter Rows:

	user	host
▶	departments_user	localhost
	mysql.infoschema	localhost
	mysql.session	localhost
	mysql.sys	localhost
	root	localhost
	user1	localhost

11. Show privileges assigned to a specific user.

```
35 •  SHOW GRANTS FOR 'user1'@'localhost';
```

OUTPUT:

Result Grid Filter Rows: Ex

	Grants for user1@localhost
▶	GRANT USAGE ON *.* TO `user1`@`localhost`
	GRANT SELECT ON `labsheet4_db`.`student` ...